

# Curriculum Vitae - Prateek Dwivedi

## Education

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| <b>Indian Institute of Science Education and Research (IISER), Pune</b><br><i>BS-MS Dual Degree: Mathematics</i>                                                                             | 2020 – 2025    |
| <b>Technische Universität (TU) Ilmenau, Germany</b><br><i>Visiting Research: Mathematics (Coding Theory)</i>                                                                                 | 2024 – 2025    |
| <b>IIIT Delhi</b><br><i>PhD: Mathematics (Combinatorics); discontinued</i>                                                                                                                   | 2025           |
| <b>IIT Madras</b><br><i>PhD (School of Interdisciplinary Studies - SIDiS) under the joint supervision of Prof. S Ganga Prasath, Interface Lab and Prof. Sayan Gupta, The Uncertainty Lab</i> | 2026 – Present |

## Research Experience

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| <b>Statistical Modelling of Gene Expression</b><br><i>IISER Pune (Prof. Leelavati Narlikar)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2022 |
| <ul style="list-style-type: none"><li>Built statistical models to predict gene expression from promoter sequence data</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |
| <b>Stochastic Processes: Martingale Reading Project</b><br><i>IISER Pune (Prof. Anup Biswas)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2023 |
| <ul style="list-style-type: none"><li>Read an introduction to martingale theory from the book “Stochastic Calculus and Applications” by Cohen and Elliott, and an alternative interpretation of Doob’s Martingale Inequality from “A Trajectorial Interpretation of Doob’s Martingale Inequalities” by Acciaio, Beiglböck, Penkner, Schachermayer, and Temme.</li></ul>                                                                                                                                                                                                                                                             |      |
| <b>Coding Theory Project: Bounds on Asymptotic Rate Function</b><br><i>IISER Pune (Prof. Krishna Kaipa)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2023 |
| <ul style="list-style-type: none"><li>Studied bounds on asymptotic rate function and produced alternative proofs of the MacWilliams identity for block codes</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |
| <b>Finite Geometry: Classification of Conic Pencils</b><br><i>IISER Pune (Prof. Krishna Kaipa)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2024 |
| <ul style="list-style-type: none"><li>Classified pencils of conics in <math>\text{PG}(2, q)</math> under the action of <math>\text{PGL}_3(q)</math></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |
| <b>Convolutional Coding Theory: Thesis Research</b><br><i>TU Ilmenau (Prof. Julia Lieb) &amp; IISER Pune (Prof. Krishna Kaipa)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2024 |
| <ul style="list-style-type: none"><li><b>Title:</b> “Convolutional Codes: MacWilliams Identity and Hulls”</li><li>Read papers on the MacWilliams identity for convolutional codes by Prof. Heide Gluesing-Luerssen and Gert Schneider and produced a literature review</li><li><b>Original contribution:</b> extended LCD codes and code hulls to the convolutional setting and proved a hull-dimension formula using the rank of the generator matrix, and gave an explicit construction for convolutional LCD codes using Lagrange’s four-square theorem, as an extension to a similar construction for block LCD codes</li></ul> |      |
| <b>Topological Combinatorics: Arrangements of Curves</b><br><i>IIIT Delhi (Prof. Rajiv Raman)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025 |
| <ul style="list-style-type: none"><li>Investigating problems in topological combinatorics related to arrangements of curves and Levi’s Extension Theorem for non-piercing curves</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |
| <b>Interpretable Attention Models for Dynamics</b><br><i>IIT Madras (Prof. S Ganga Prasath &amp; Prof. Sayan Gupta)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026 |

## Additional Information

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- Speaker, Coding Theory: SURC 2023 (Summer Undergraduate Research Conference): Azim Premji University
- Attendee, Indo-European Conference on Mathematics (IECM) January 2026, IISER Pune and SPPU